THE JOURNAL

ORIGINAL ARTICLES

Methods

otherwise eligible for inclusion were excluded from analyses because they were missing data on one or more of the study measures (n = 707). The study participants were 749 5th, 761 7th, and 479 9th graders, comprising 1012 boys and 977 girls attending a middle-to-high income Southern California school district in 2002 to 2003. The demographic characteristics of the 1989 students included in the analyses resembled the 2696 who were eligible (**Table I**). Fifth graders comprised 37.7% of the analytic sample but only 31% of the population, indicating some over-representation of 5th graders. Ninth graders comprised 24.1% of the analytic sample but 32.5% of the population, indicating some under-representation of 9th graders. Departures from representativeness involving ethnicity and sex were negligible, with differences in proportions of ethnic compositnic tht analyti491344ple ot370tde52(o-1.1dJsia244(seg)-3taki491Jsiafrness)m%

for-age percentiles, LMS parameters were provided by the CDC[7] and were used to generate Z-scores of BMI values that were then applied to estimate sex-specific BMI-for-age percentiles. Obesity risk classification was determined for each student. We opted to use the CDC weight status cut-points,[8,9] in which the 85th to 94th percentile category is termed "overweight" and the 95th+ percentile is termed "obese." Recent policy statements now speak of the "obese child" rather than limiting the term "obese" to adults.[10] Hence, BMI percentile categories included <5th, ≥5th to 84th, ≥85th to 94th, and ≥95th, which correspond to the folperformance. African American students were less likely to achieve California fitness standards than Asian American and non-Hispanic white students (OR_{Asians} = .49, P = .04; OR_{whites} = .58, P = .005).

Obesity status and mean BMI-for-age percentiles (converted from z-scores for ease of interpretation) are also included in **Table II**. The combined prevalence of overweight and obese (BMI ≥85th percentile) was 31.8% for boys and 27.7% for girls (**Table II**). Mean BMI-for-age percentiles varied by ethnicity, with 16.3% of Asians classified as either at risk for overweight or obesity, 22.1% of non-Hispanic whites, 40.6% of African Americans, and 47.7% of Hispanics. The percentages of students classified as overweight or obese was greater among African Americans and Hispanics than among Asians or non-Hispanic whites (all comparisons $P < .003$, after Bonferroni correction). Additionally, Asian/Pacific Islanders had slightly lower BMI-for-age percentiles than non-Hispanic whites ($F(1,11) = 10.65$, $P = .04$). Hispanics and African Americans did not differ significantly from each other, nor did boys' BMI-for-age differ on average from that of girls.

With respect to age-specific fitness standards, students who failed to run the mile in the appropriate time interval established as appropriate for each age and sex scored significantly lower on the CAT6 and CST math, reading, and language California standards tests compared with those students who fell in the healthy fitness zone (**Table III**). Tests for linear trends revealed that decreasing quintiles of aerobic fitness scored progressively lower on CAT6 math and reading (linear trend, $P_{math} < .0001$; $P_{reading} = .001$) (**Figure 1**) and on CST math and language tests (linear trend, $P_{math} < .0001$; $P_{language} < .0001$) (**Figure 2**; available at www.jpeds.com).

Table III depicts test scores by BMI percentiles; as observed for mile run/walk time, those who exceeded both the 85th and 95th -169(ages)TJtPd 95thlin82(I-for-ag)-10(e)-346(peth and explained 15.9% of the null model variance. Adding BMI-for-age to the demographic variables-augmented model decreased the null model variance only an additional 0.4%, although this change was still statistically significant (model difference likelihood ratio ^{2} (1) = 8.6, P = .003). The full model, including all of the foregoing covariates/ predictors (including BMI-for-age) but also including a measure of the student's performance on the 1-mile fitness test decreased the null model variance an additional 3.5% (model difference likelihood ratio ^{2} (1) = 81.6, $P < .0001$). With the inclusion of the fitness measure, the student's BMI-for-age was no longer a significant contributor to the CAT6 2002 math test

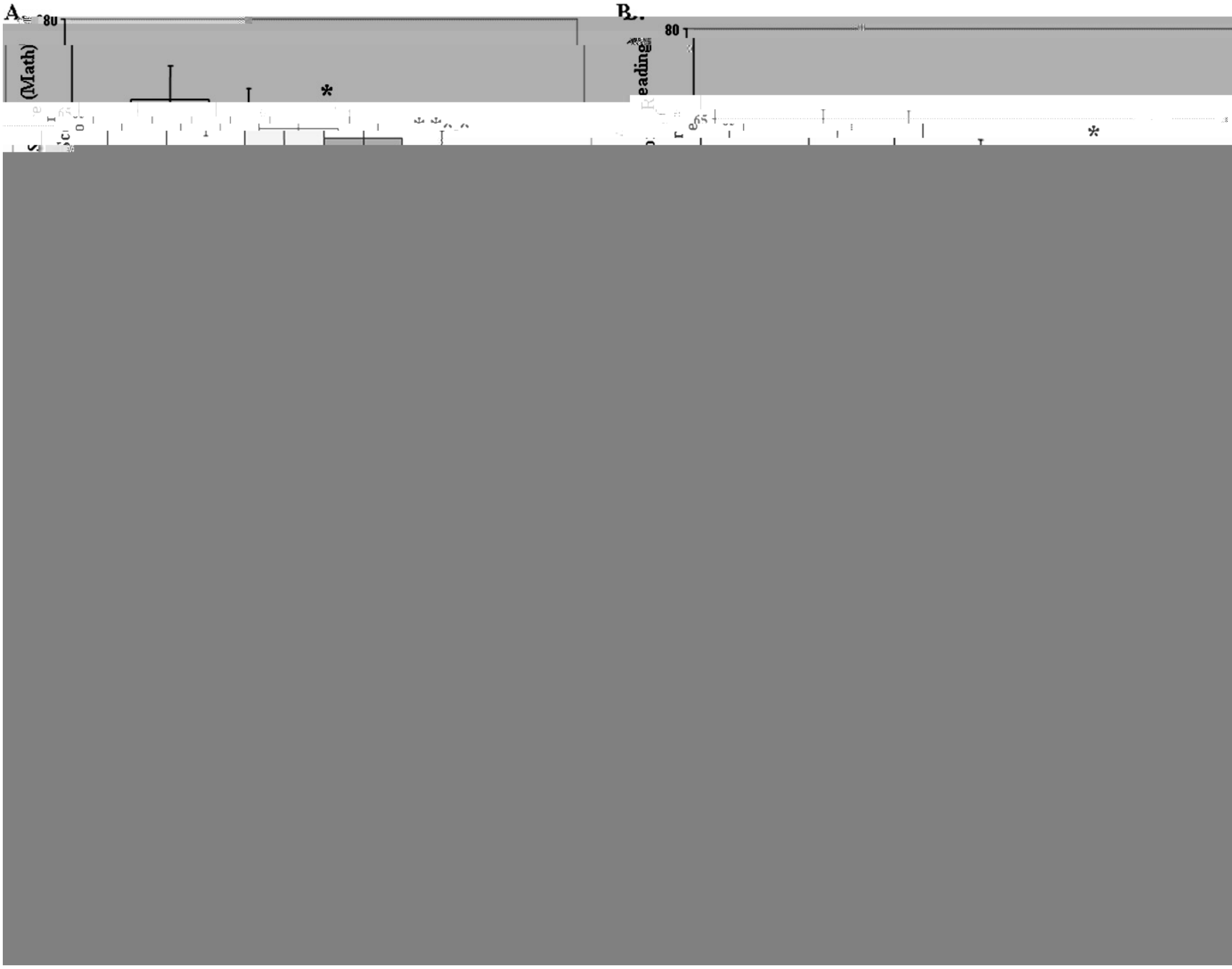

Figure 1. CAT6 mean scores, adjusted by the covariates sex, free and reduced-price lunch eligibility, and ethnicity. **A,** Percentiles in CAT6 score (math) by quintiles of minutes to complete mile run; *3rd quintile of mile time differed on 2002 CAT6 math score from 1st quintile, z = −3.33, P = .001; **4th quintile of mile time differed on 2002 CAT6 math score from 1st quintile, z = −5.86, P < .0001; **5th quintile of mile time differed on 2002 CAT6 math score from 1st quintile, z = −8.41, P < .0001. **B,** Percentile in CAT6 score (language) by quintiles of minutes to complete mile run. *4th quintile of mile time differed on 2002 CAT6 language score from 1st quintile, z = −2.65, P = .008; **5th quintile of mile time differed on 2002 CAT6 language score from 1st quintile, z = −5.07, P < .0001. **C,** Percentiles in CAT6 score (math) by BMI quintile; *4th quintile of BMI differed on 2002 CAT6 math score from 1st quintile, z = −2.40, P = .016; **5th quintile of BMI differed on 2002 CAT6 math score from 1st quintile, z = −4.02, P <.0001. **D,** Percentiles in CAT6 score (language) by BMI quintile. *5th quintile of BMI differed on 2002 CAT6 language score from 1st quintile, z = −2.35, P = .019. Error bars are 95% confidence intervals.

decreased socioeconomic status has been consistently associated with decreased standardized test scores,[12] it is an obvious potential confounder of any association between obesity and standardized test performance. Controlling for age, socioeconomic status, sex, and ethnicity did attenuate the significance of the relationships between aerobic fitness and standardized test scores and between BMI-for-age percentiles and standardized test scores. Nevertheless, associations remained significant. It appears that both BMI-for-age and performance on the 1-mile run/walk predict standardized test score performance, above and beyond the large amount of variance predicted by sex, ethnicity, and socioeconomic status, but the remaining variance explained is small.

The findings presented here confirm and extend previous findings that aerobic fitness is associated with enhanced performance on standardized achievement tests. The extensions include generalization to ethnically and socioeconomically diverse students varying across primary and secondary school grades, using objective measures of both aerobic and standardized test score performance. Additionally, the data suggest that the association of obesity status and test score performance may be mediated by fitness.

Our data indicate that the fitness level of nearly two thirds of the students surveyed did not fall in the healthy fitness zone. These data are in agreement with the 2004 California Fitness Test data, in which only 27% of more than 1.3 million students tested in grades 5, 7, and 9 met fitness standards in all 6 examined variables. Decades of declining participation in physical education[13-15] have resulted in decades of declining student fitness levels.[16] Sixty-one percent of children ages 9 to 13 years do not participate in any organized physical activity during their nonschool hours.[17] The result is that one third of adolescents fail to achieve a recommended minimum of 30 minutes of moderate to vigorous physical activity 3 times per week.[18,19]

We also noted ethnic disparities in aerobic fitness, with both Hispanic and African American youth reporting slower mile run/walk performances than whites and Asians. This finding is consistent with that of Beets et al,[20] who reported a similar trend in all California students in 2002. Participation in vigorous activity is higher in whites (67%) compared with blacks (54%) and Hispanics (60%) and decreases with advancing grade. The higher BMI-for-age (**Table II**) presenting in African Americans and Hispanics may also contribute to their lower aerobic fitness levels.

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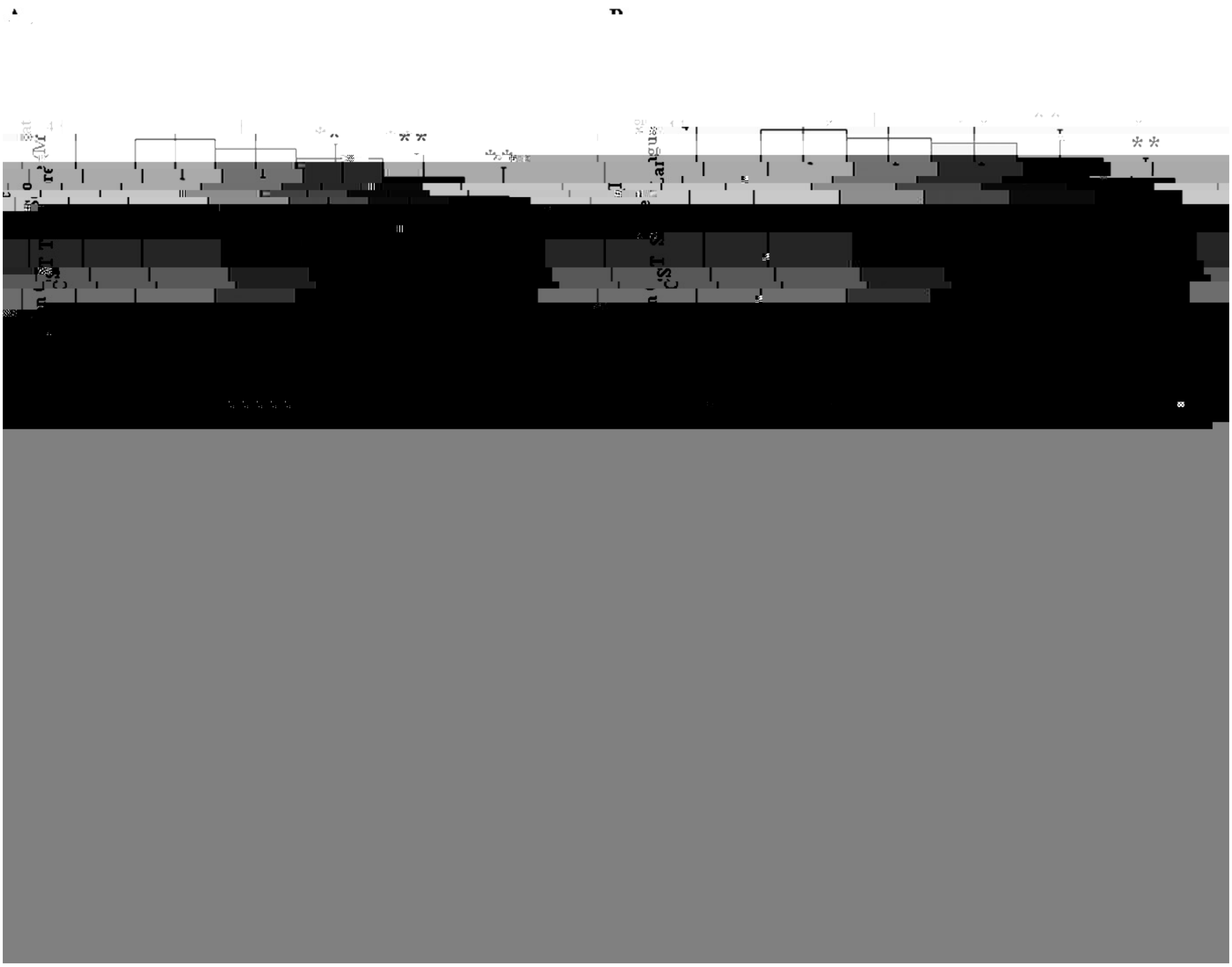

Figure 2. CST mean scores, adjusted by the covariates sex, free and reduced-price lunch eligibility, and ethnicity. **A,** Percentiles in CST score (math) by quintiles of minutes to complete mile run; *3rd quintile of mile time differed on 2002 CST math score from 1st quintile, $z = -2.27$, $P = .023$; **4th quintile of mile time differed on 2002 CST math score from 1st quintile, $z = -5.22$, $P < .0001$; **5th quintile of mile time differed on 2002 CST math score from 1st quintile, $z = -8.11$, $P < .0001$. **B,** Percentile in CST score (language) by quintiles of minutes to complete mile run; **4th quintile of mile time differed on 2002 CST language score from 1st quintile, $z = -3.41$, $P < .0001$; **5th quintile of mile time differed on 2002 CST language score from 1st quintile, $z = -6.46$, $P < .0001$. **C,** Percentiles in CST score (math) by BMI quintile; **5th quintile of BMI differed on 2002 CST math score from 1st quintile, $z = -5.14$, $P <.0001$. **D,** Percentiles in CST score (language) by BMI quintile; *5th quintile of BMI differed on 2002 CST language score from 1st quintile, $z = -2.51$, $P = .01$. Error bars are 95% confidence intervals.